

CSC 535 — Homework 1

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Problem 1

We have a random sample of 10 Bernoulli trials with success probability ($p = 0.5$).

Let

$$X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(0.5), \quad i = 1, \dots, 10$$

a)

The number of heads is equal to the number of tails when the number of successes is 5:

$$\sum_{i=1}^{10} X_i = 5$$

The probability of this event is given by the binomial formula:

$$\mathbb{P} \left(\sum_{i=1}^{10} X_i = 5 \right) = \binom{10}{5} \left(\frac{1}{2} \right)^{10}$$

b)

The number of heads is greater than the number of tails when the number of successes is greater than 5:

$$\sum_{i=1}^{10} X_i > 5$$

The probability of this event is given by the binomial formula:

$$\mathbb{P} \left(\sum_{i=1}^{10} X_i > 5 \right) = \sum_{x=6}^{10} \binom{10}{x} \left(\frac{1}{2} \right)^{10}$$

c)

The result of i th flip is symmetric with $11 - i$ th flip is:

$$X_i = X_{11-i}, \quad i = 1, 2, 3, 4, 5$$

The probability is given by the independent property:

$$\mathbb{P}(X_i = X_{11-i}, \quad i = 1, 2, 3, 4, 5) = \prod_{i=1}^5 \mathbb{P}(X_i = X_{11-i})$$

By the fact that X_i, X_{11-i} follows an identical distribution for $i = 1, \dots, 5$:

$$\mathbb{P}(X_i = X_{11-i}) = \mathbb{P}(X_1 = X_2 = 1 \text{ or } X_1 = X_2 = 0) = \mathbb{P}(X_1 = X_2 = 1) + \mathbb{P}(X_1 = X_2 = 0)$$

Therefore,

$$\mathbb{P}(X_i = X_{11-i}, \quad i = 1, 2, 3, 4, 5) = \left(\frac{1}{2}\right)^5$$

d)

We define $a(n)$ as the number of binary sequences of length n that do not contain 4 or more consecutive heads.

Since each sequence $(X_1, \dots, X_{10})$ has equal probability, the probability of seeing at least four consecutive heads is:

$$\mathbb{P}(\text{at least 4 consecutive heads in } (X_1, \dots, X_{10})) = 1 - \mathbb{P}(\text{not contain 4 or more consecutive heads})$$

To define $a(n)$ recursively, we consider how a valid sequence of length n (i.e., no 4 consecutive heads) can be constructed:

- If the sequence ends in a tail (T), we can append it to any valid sequence of length $n - 1$, contributing $a(n - 1)$ possibilities.
- If the sequence ends in one, two, or three heads (H), it must be preceded by a tail. So we can append:
 - TH to any valid sequence of length $n - 1$: contributes $a(n - 2)$
 - THH to any valid sequence of length $n - 2$: contributes $a(n - 3)$
 - $THHH$ to any valid sequence of length $n - 3$: contributes $a(n - 4)$

This leads to the recurrence relation:

$$a(n) = a(n - 1) + a(n - 2) + a(n - 3) + a(n - 4)$$

with base cases:

$$\begin{aligned}a(0) &= 1 && \text{(empty sequence)} \\a(1) &= 2 && \text{(H, T)} \\a(2) &= 4 && \text{(all 2-flip sequences valid)} \\a(3) &= 8 && \text{(all 3-flip sequences valid)}\end{aligned}$$

We can now compute $a(10)$ numerically.

```
In [32]: # Count number of sequences with no 4 consecutive heads
def count_valid_sequences(n):
    a = [0] * (n + 1)
    a[0] = 1 # empty sequence
    if n >= 1: a[1] = 2
    if n >= 2: a[2] = 4
    if n >= 3: a[3] = 8
    if n >= 4: a[4] = 15 # all except 'HHHH'
    for i in range(5, n+1):
        a[i] = a[i-1] + a[i-2] + a[i-3] + a[i-4]
    return a[n]

# Total possible sequences
n = 10
total = 2 ** n
valid = count_valid_sequences(n)
at_least_4_heads = total - valid
probability = at_least_4_heads / total

print(f"Number of sequences with at least 4 consecutive heads: {at_least_4_h
print(f"Probability: {probability:.5f}")
```

```
Number of sequences with at least 4 consecutive heads: 251
Probability: 0.24512
```

Problem 2

a)

Let Ω be the sample space for rolling two fair six-sided dice, and let $X_1(\omega)$ and $X_2(\omega)$ denote the outcomes of the first and second die, respectively, where $X_1, X_2 \in \{1, \dots, 6\}$.

Since each die has 6 possible outcomes and the rolls are independent, the total number of possible outcomes is:

$$\#\{X_1(\omega) : \omega \in \Omega\} \times \#\{X_2(\omega) : \omega \in \Omega\} = 6 \times 6 = 36$$

There is only 1 outcome. The exact probability is:

$$\mathbb{P}(X_1 + X_2 = 2) = \frac{1}{36} \approx 0.02778$$

b)

```
In [33]: import numpy as np

# Set seed
np.random.seed(0)

# Simulate 1000 throws of two 6-sided dice
die1 = np.random.randint(1, 7, 1000)
die2 = np.random.randint(1, 7, 1000)
sums = die1 + die2

# Calculate empirical probability
empirical_prob = np.mean(sums == 2)
print(f"Empirical probability of sum = 2: {empirical_prob:.5f}")
```

Empirical probability of sum = 2: 0.02900

c)

```
In [34]: # Repeat simulation 10 times with the same seed
results = []

for i in range(10):
    np.random.seed(i)
    die1 = np.random.randint(1, 7, 1000)
    die2 = np.random.randint(1, 7, 1000)
    sums = die1 + die2
    prob = np.mean(sums == 2)
    results.append(prob)

# Print results
for i, prob in enumerate(results, 1):
    print(f"Run {i}: Empirical probability of sum = 2: {prob:.5f}")
```

Run 1: Empirical probability of sum = 2: 0.02900
Run 2: Empirical probability of sum = 2: 0.02400
Run 3: Empirical probability of sum = 2: 0.02000
Run 4: Empirical probability of sum = 2: 0.02300
Run 5: Empirical probability of sum = 2: 0.03400
Run 6: Empirical probability of sum = 2: 0.03200
Run 7: Empirical probability of sum = 2: 0.02800
Run 8: Empirical probability of sum = 2: 0.04000
Run 9: Empirical probability of sum = 2: 0.01900
Run 10: Empirical probability of sum = 2: 0.03400

d)

Since the random sample consists of finite observations, the empirical probability will vary from one simulation to another, while the true probability of the event remains fixed and constant.

Problem 3

a)

Since the event $(A = 3, B = 3)$ is one of the outcomes contributing to the event $(S = 6)$, where $(S = A + B)$, and $(S = 6)$ includes outcomes of the form $(A = i, B = 6 - i)$ for $(i = 1, 2, 3, 4, 5)$, we have:

$$\mathbb{P}(A = 3, B = 3 \mid S = 6) = \frac{\mathbb{P}(A = 3, B = 3, S = 6)}{\mathbb{P}(S = 6)} = \frac{\mathbb{P}(A = 3, B = 3)}{\mathbb{P}(S = 6)} = \frac{\mathbb{P}(A = 3)}{\mathbb{P}(S = 6)}$$

b)

Since $(A = 3, B = 3)$ implies $(S = 6)$, we have:

$$\mathbb{P}(S = 6 \mid A = 3, B = 3) = \frac{\mathbb{P}(A = 3, B = 3, S = 6)}{\mathbb{P}(A = 3, B = 3)} = \frac{\mathbb{P}(A = 3, B = 3)}{\mathbb{P}(A = 3, B = 3)} = 1$$

c)

The event $(A = 1, B = 1)$ is mutually exclusive with $(S = 6)$, since $(1 + 1 \neq 6)$. Therefore:

$$\mathbb{P}(A = 1, B = 1 \mid S = 6) = \frac{\mathbb{P}(A = 1, B = 1, S = 6)}{\mathbb{P}(S = 6)} = \frac{0}{\mathbb{P}(S = 6)} = 0$$

d)

The event $(S = 2)$ implies $(A = 1)$, since it is the only possible pair that sums to 2. Therefore:

$$\mathbb{P}(A = 1 \mid S = 2) = \frac{\mathbb{P}(A = 1, S = 2)}{\mathbb{P}(S = 2)} = \frac{\mathbb{P}(S = 2)}{\mathbb{P}(S = 2)} = 1$$

e)

The total number of outcomes with $(A + B = 6)$ is 5, out of 36 total possible dice rolls. So:

$$\mathbb{P}(S = 6) = \frac{5}{36}$$

Problem 4

a)

We define the gaussian pdf:

$$\phi(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

, where $\mu = 70, \sigma = 2$.

```
In [35]: import matplotlib.pyplot as plt

# Parameters
mu = 70
sigma = 2

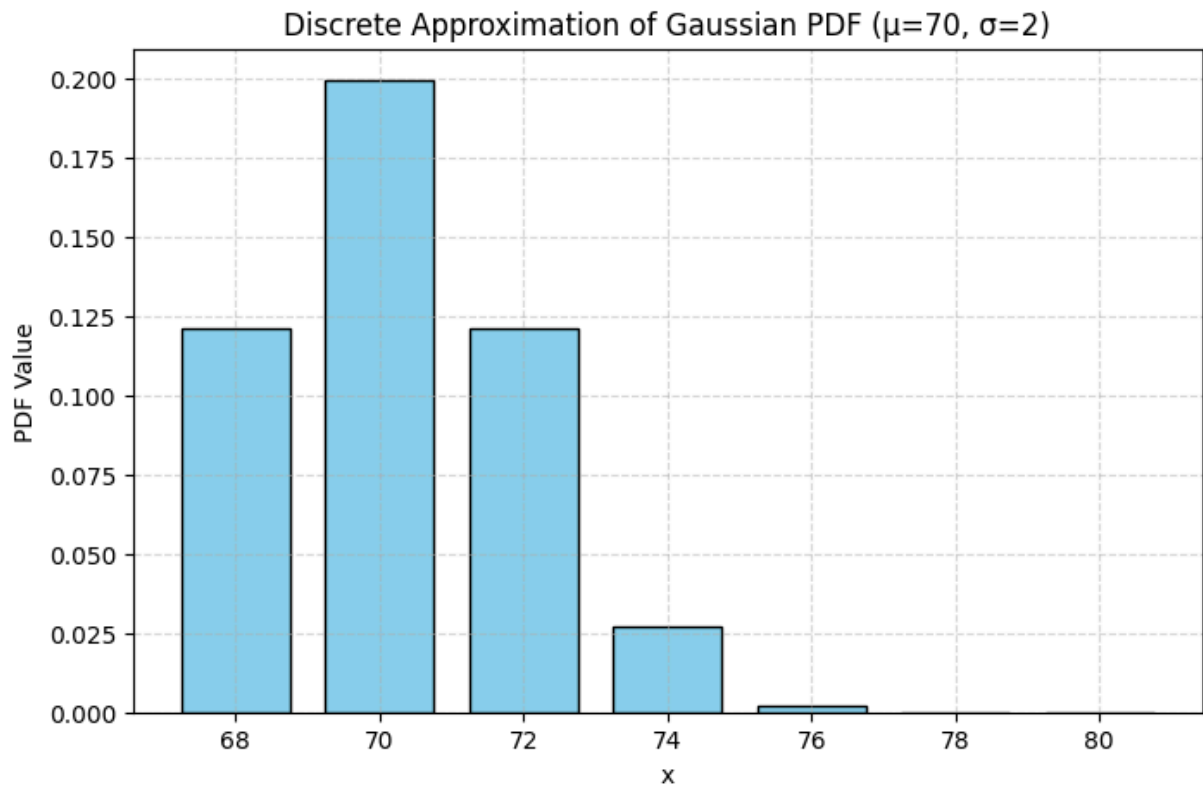
# Generate an array x
delta_x = 2 # increment of the array x
x = np.arange(68, 80+delta_x, delta_x)

# Gaussian PDF
def gaussian_pdf(x, mu, sigma):
    coef = 1 / (np.sqrt(2 * np.pi)*sigma)
    exponent = -((x-mu)**2) / (2 * sigma**2)
    return coef * np.exp(exponent)

# Evaluate the PDF at each x
p = gaussian_pdf(x, mu, sigma)

# Plot as a bar chart
plt.figure(figsize=(8, 5))

plt.bar(x, p, width=1.5, color="skyblue", edgecolor="black")
plt.title("Discrete Approximation of Gaussian PDF (μ=70, σ=2)")
plt.xlabel("x")
plt.ylabel("PDF Value")
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()
```



b)

We find an integral:

$$\mathbb{P}(68 < X \leq 80) = \int_{68}^{80} \phi(x|\mu, \sigma^2) dx$$

The Riemann sum approximation is defined by:

$$\sum_{i=1}^N \phi(x_i|\mu, \sigma^2) \delta(x)$$

, where $N = \#\{x_i\}$ and $\delta x = \max_{1 \leq i \leq N-1} |x_{i+1} - x_i|$.

```
In [36]: # Find an integral using a Riemann sum.

integral_approx = np.sum(gaussian_pdf(x, mu, sigma)*delta_x)

print(f"Integral approximation with the delta x = 2 : {integral_approx:.5f}")
```

Integral approximation with the delta x = 2 : 0.94144

c)

We reduce the space with $\delta x = 0.01$.

```
In [37]: # Generate a finer grid for x with smaller spacing
delta_x_fine = 0.01
x_fine = np.arange(68, 80 + delta_x_fine, delta_x_fine)

# Approximate the integral of the Gaussian PDF over [68, 80]
integral_approx = np.sum(gaussian_pdf(x_fine, mu, sigma) * delta_x_fine)

print(f"Integral approximation over [68, 80] with  $\Delta x = \{delta\_x\_fine\}$ : {integ
```

Integral approximation over [68, 80] with $\Delta x = 0.01$: 0.84195

The integral approximation has decreased compared to the previous coarser estimate. While the finer discretization improves accuracy, it also comes with the practical downside of increased computational cost.

d)

```
In [38]: x_wide = np.arange(20, 120 + delta_x_fine, delta_x_fine)

# PDF
p_wide = gaussian_pdf(x_wide, mu, sigma)

# Approximate the integral of the Gaussian PDF over [68, 80]
integral_approx = np.sum(p_wide * delta_x_fine)

# Print the approximation result
print(f"Approximate probability  $P(20 \leq X < 120)$ : {integral_approx:.10f}")
```

Approximate probability $P(20 \leq X < 120)$: 1.0000000000